

# Gender shades: Intersectional accuracy disparities in commercial gender classification (Buolamwini & Gebru, 2018)

## Read the whole thing

## Reading Notes

bias in AI and/or algorithms can have impacts beyond the immediate context in which that bias appears

introduces a facial dataset that's more balanced than those that came before

darker female faces miscategorized most often (female more than male, darker more than lighter)

## Reading Questions

1. Buolamwini & Gebru use the Fitzpatrick skin type scale as a measurement instrument. What are its limitations for this study? How does measurement instrument choice become a source of bias in the study itself — and what does that imply for how we interpret the findings?

As they mention in the article, the Fitzpatrick system is somewhat coarse in their view since it allocates three points on the scale to participants who would be perceived as “white” and three to those would be perceived otherwise. The use of this instrument could then become a source of bias by not providing a suitably fine-grained method of classification, which would then mean that the findings could be compromised. However, given that the authors intentionally chose relatively discrete differences in phenotype, this is likely not a concern for this study. It would be, however, were they to include more faces which cluster toward the middle of the Fitzpatrick scale.

2. The study found that **darker-skinned females** showed the largest accuracy disparity across all three commercial systems tested. What combination of representation bias (in training data) and measurement bias (in the Fitzpatrick scale) could account for this specific intersectional pattern?

The likely source of representation bias is that the training data included a larger/disproportionate number of male faces and lighter/whiter faces. Also, given

the coarseness of the Fitzpatrick scale that Buolamwini and Gebru (2018) take issue with, the information they were able to encode in the PPB might have been insufficiently detailed to allow for models to learn.

3. The paper was published in 2018. IBM, Microsoft, and Face++ were notified of the findings. What remediation steps did these companies take? Were those steps sufficient to address the root causes identified in the paper? (Note: subsequent audits by researchers including the ACLU in 2018 and 2019 found continued disparities.)

The article does not mention any remediation steps. However, if later audits found continued disparities, then more work is necessary.

4. If a university deployed a facial recognition system for building access management, what populations would be most at risk based on the Gender Shades findings? What governance steps would be required before such a deployment?

Darker females would be more at risk. Were the university interested in deploying such a technology, they would need to ensure that the system they wanted to use would be able to accurately recognize those who need access. This would be particularly challenging since differing clothes, eyewear, headwear, makeup, and lighting could all play roles. If this might, for instance, deny students access to dormitories or employees to their workplaces, this technology could easily create more problems than it solves.

## Key Terms

| Term                        | My Definition                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intersectional bias         | the term is not used in the paper, but it refers to unequal treatment or judgment applied to someone as a result of holding multiple identities (e.g., age, race, socioeconomic status, sexual orientation, etc.) |
| Fitzpatrick skin type scale | a six-item scale used in human dermatology for classifying the color of skin                                                                                                                                      |
| Representation bias         | the term is not used in the paper, but it refers to a lack of diversity in the training data used for an AI model                                                                                                 |
| Benchmark dataset bias      | the term is not used in the paper, but it refers to systemic errors or prejudices that exist in AI models, typically as a result of training data                                                                 |

|                    |                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Accuracy disparity | the term is not used in the paper, but it refers to models performing better for certain demographic groups/subgroups than others |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|

## On the dangers of stochastic parrots: Can language models be too big? (Bender et al., 2021)

**Read Abstract, Section 1, Section 5, and Section 6**

### Reading Notes

essentially, the authors are asking whether bigger is always better and arguing for (1) more judicious choices in what data models are trained on and (2) that environmental and financial concerns should come first

they argue that LLM-generated content is incapable of being coherent because coherence is predicated on two people communicating in such a way that they are considering each other's mental states

the problem with LLMs is if the content is presented as meaningful, then we're conditioned to accept it as meaningful, even though it might be nonsense

the existence of bias within LLMs risks compounding and perpetuating that bias, especially if generated content is then used to train future models

### Reading Questions

1. What is the 'stochastic parrot' metaphor and what precise claim does it make about the relationship between LLM-generated text and meaning or understanding? Do you find this framing accurate, overstated, or understated?

The stochastic parrot metaphor means that LLMs are probabilistic in nature and do not “understand” what they are generating in the way that humans do. I find this framing to be somewhat overstated because while in a certain sense it is true, if these stochastic generations are frequently accurate, it seems immaterial to me whether the model “understands” what it is generating or not. These stochastic processes are also at work in humans. If you were give someone a sentence and ask them to predict the next word, you could probably get fairly close to an LLM probability if you polled enough people (wisdom of the crowd).

2. Section 6.2 argues that LLM training on internet text encodes and amplifies existing social biases because internet text over-represents certain demographics and

viewpoints. How does this argument connect to the Gender Shades findings from Reading 1?

Both articles discussed how skewed aspects of data used to train AI models can result in skewed results. For example, if certain groups or views are over-represented in training data, then you can reasonably expect that output data will do the same.

3. The paper was published at FAccT 2021 and generated significant controversy, partly because it led to a researcher's departure from Google. Does the paper's institutional context (two of its four authors were at Google; one was asked to retract or remove her name) affect how you read its arguments? Why or why not?

It might have a small effect, but I'd need to know more. Just because some of the researchers were at a company that works heavily in AI doesn't mean they were themselves deeply involved in AI. And even if they were, I'd want to be a bit cautious of saying that because they work(ed) at Google, that somehow makes their perspective more valid than if it were written by an academic or a concerned citizen.

4. Wei et al. (2022) — which you read in Module 2 — argue that emergent reasoning capabilities in LLMs suggest something beyond pure pattern-matching. Does this finding challenge, complement, or leave unaddressed the Bender et al. 'stochastic parrot' claim? Explain your reasoning.

The Wei et al. (2022) finding challenges the Bender et al. (2021) finding because they're making opposite claims: the former think that LLMs may be doing something more than just probabilistic prediction, while the latter argue that LLMs are nothing more than unthinking prediction machines. These are incompatible claims.

## Key Terms

| Term                            | My Definition                                                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stochastic parrot               | a term for LLM which refers to their ability to create seemingly coherent text/output by using a probabilistic engine                                                                             |
| Bias encoding in training data  | prejudice (whether overt or covert) which becomes a part of an LLM as a result of it existing in the training data                                                                                |
| Misdirected research efforts    | a misdirected research effort would be one in which effort is put toward a research goal which is not worth the effort                                                                            |
| Coherence without comprehension | the ability of an LLM to generate content which makes sense and is understandable to a human reader/listener without the AI/LLM actually understanding that content in the way that a human would |

